

## $k$ -means Clustering & $k$ -means++

- Given  $\mathcal{X} = \{x_1, \dots, x_n\} \subset \mathbf{R}^D$  and integer  $k$ , find  $k$  centers  $C$  minimizing

$$\text{cost}(\mathcal{X}, C) = \sum_{x \in \mathcal{X}} \min_{c \in C} \|x - c\|^2$$

- NP-hard** in general; practical heuristic is **Lloyd's iterations** (alternating assignment and centroid update). Lloyd's needs a good **initialization (seeding)** to avoid poor local minima
- Choose  $c_1 \in \mathcal{X}$  uniformly. Given centers  $C_t$ , sample next center via the  $D^2$  distribution:

$$\Pr[c_{t+1} = x \mid C_t] = \frac{\text{cost}(x, C_t)}{\text{cost}(\mathcal{X}, C_t)}$$

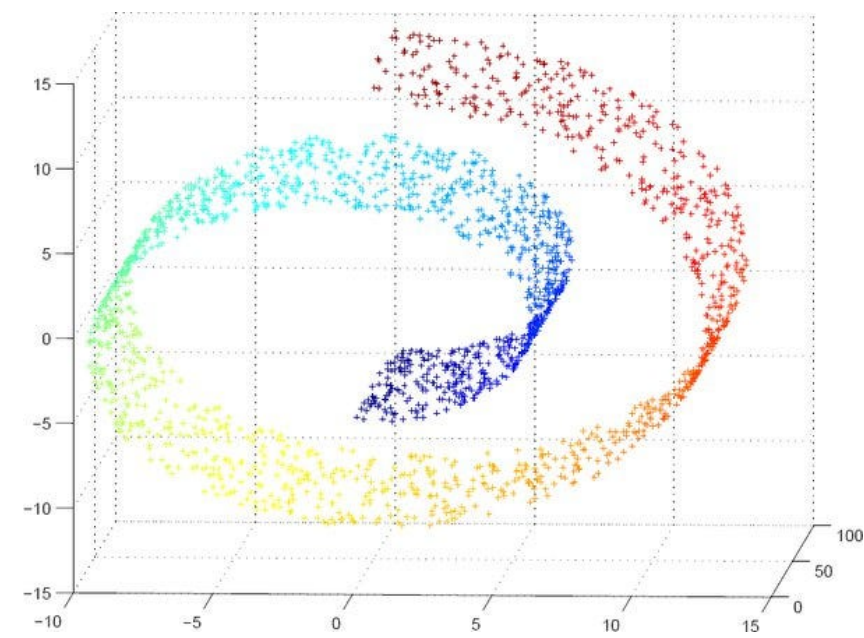
- Guarantee:**  $O(\log k)$  expected approximation (Arthur & Vassilvitskii, 2007). **Naive Runtime:**  $O(nkD)$  - each round scans all  $n$  points in  $\mathbf{R}^D$ . **Too Slow even for moderate sized data!**

## Beyond Worst Case Analysis

- Worst-case algorithms are designed to handle *any* input - including pathological ones. Can we model the *typical* inputs one actually encounters?
- Prior approaches** for  $k$ -means assume properties of the *optimal solution* -  $\alpha$ -separation, approximation stability, perturbation resilience - which are **difficult to verify** on real datasets **This work:** assume structure on the *input distribution* instead, via the manifold hypothesis

## Manifold Hypothesis

**Assumption.** Data in  $\mathbf{R}^D$  concentrates near a smooth submanifold  $\mathcal{M}$  of *intrinsic dimension*  $d \ll D$ .



- Well-established assumption underlying modern deep learning (Bengio et al., 2013; Fefferman et al., 2016)
- Assumption on the *data*, not on the optimal solution  $\Rightarrow$  easier to validate empirically

## Theorem

*Rejection sampling under manifold hypothesis* Let  $\mathcal{X} = \{x_1, \dots, x_n\} \subset \mathbf{R}^D$  be a dataset generated according to the Manifold Assumption. Then there exists an algorithm which additionally takes as input a parameter  $\rho < 1$  and outputs a set  $C$  of  $k$  centers such that

$$\mathbf{E}[\text{cost}(\mathcal{X}, C)] \in O(\rho^{-2} \log k) \text{opt}_k(\mathcal{X})$$

using a one-time preprocessing time of  $O(nD \log k)$ , with each seeding taking expected time  $\tilde{O}(\rho^{-1} k^{1+\gamma})$ , where  $\gamma = +\rho$ , and  $\tilde{O}(\cdot)$  hides  $\log k, \log n$  factors

## Scaling Laws

Define two data-dependent parameters for the dataset  $\mathcal{X}$ :

$$\beta_k(\mathcal{X}) = \frac{\text{opt}_1(\mathcal{X})}{\text{opt}_k(\mathcal{X})}, \quad \eta(\mathcal{X}) = \frac{\max_{i \neq j} \|x_i - x_j\|}{\min_{i \neq j} \|x_i - x_j\|}$$

If  $\mathcal{X}$  is sampled from a density supported on a  $d$ -dimensional manifold, then with high probability:

$$\beta_k(\mathcal{X}) \approx k^{2/d}, \quad \eta(\mathcal{X}) \approx n^{3/d}$$

We perform a large-scale empirical study across images (raw pixels and embeddings), text embeddings, tabular data, sensor data etc to corroborate our theory using existing ID-Estimation tools like MLE (Levina & Bickel, 2005).

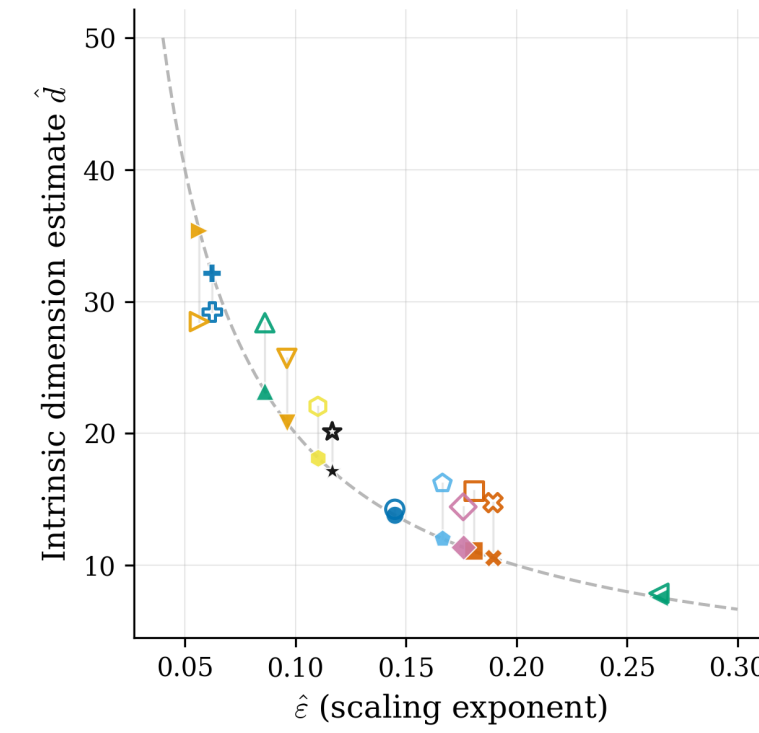


Figure:  $\epsilon$  vs  $\hat{d}_{\text{MLE}}$

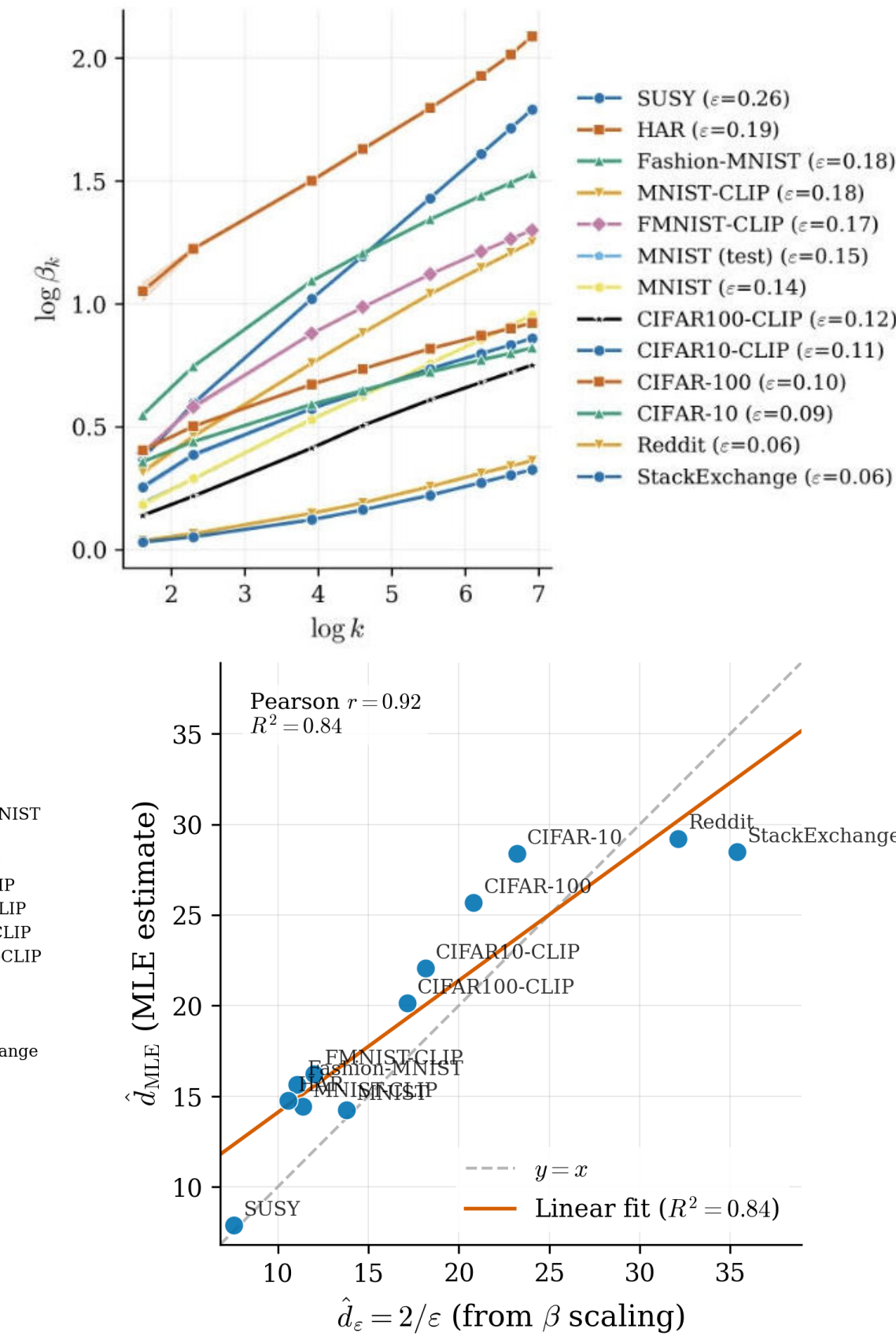


Figure:  $\hat{d}_\epsilon$  vs  $\hat{d}_{\text{MLE}}$

## Rejection Sampling for $D^2$ -Sampling

**Goal:** simulate the  $D^2$  distribution  $\pi(x \mid C) \propto \text{cost}(x, C)$  cheaply.

- Use the **proposal distribution**  $\kappa(x \mid C) \propto \|x\|^2 + \|c_1\|^2$ , **sampleable in  $O(\log n)$  via a binary tree.**
- Accept proposal  $x$  with probability  $\propto \text{cost}(x, C) / \kappa(x)$ ; the max Rényi divergence  $D_\infty(\pi \parallel \kappa) \leq \log(4\beta_k)$  controls the expected number of proposals needed.
- Use an **LSH-based ANNS** data structure to evaluate  $\text{cost}(x, C)$  approximately, avoiding the  $O(kD)$  per-point scan. **Querying and Insertion times depend on  $\log \eta(\mathcal{X})$ .**

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## Algorithm Performance

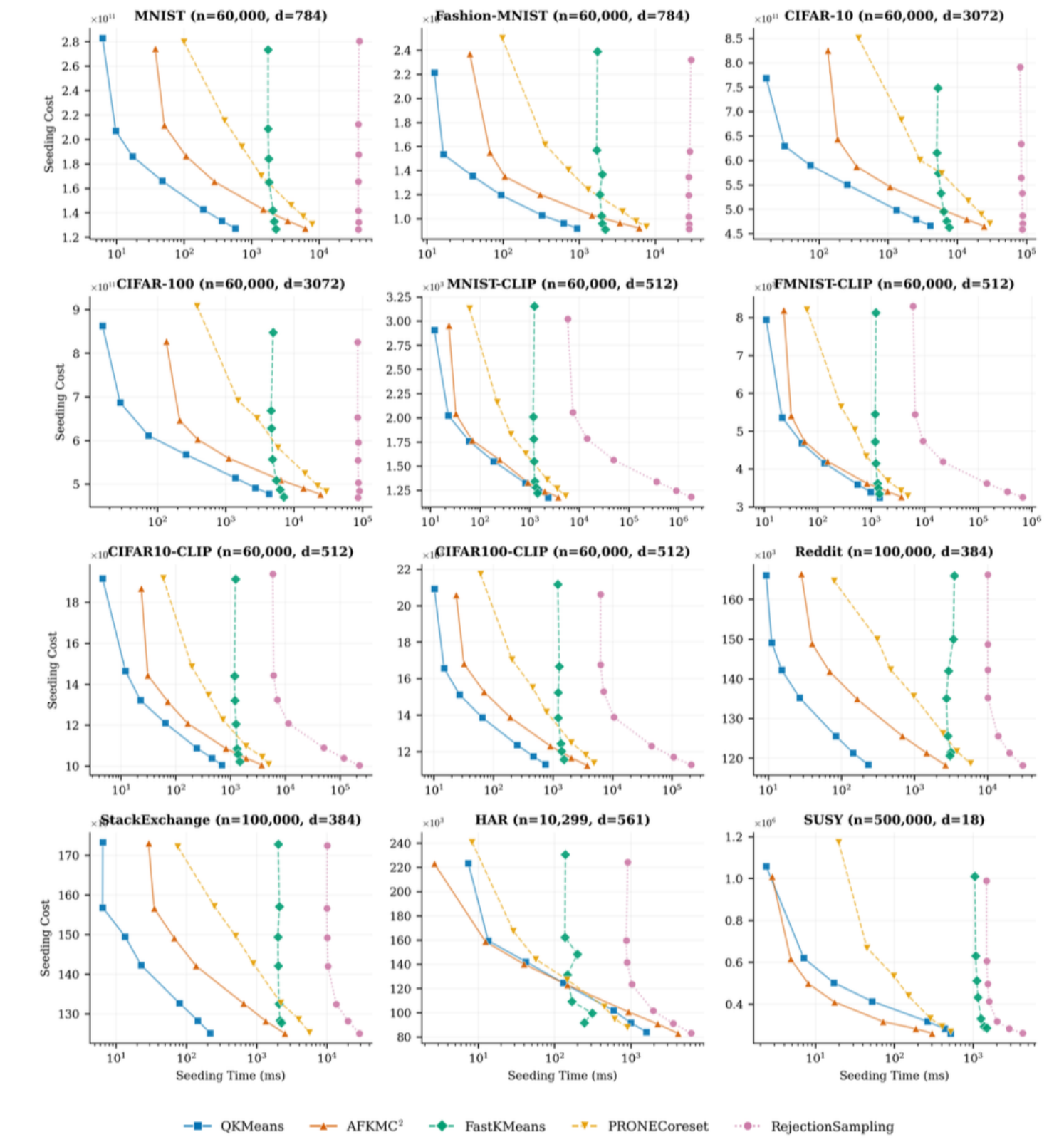
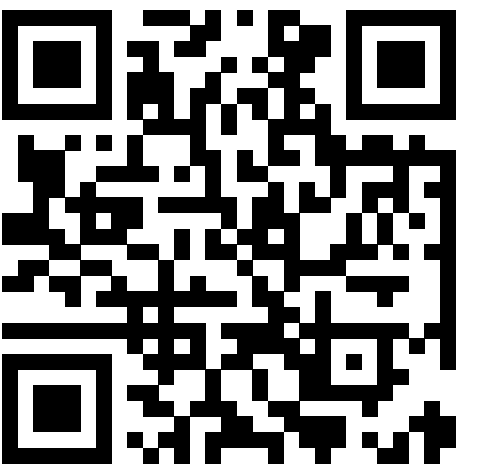


Table 1. Comparison of time complexity and approximation guarantees for accelerating  $k$ -means++ seeding. For our method, we empirically show  $\beta \sim k^\epsilon$  and  $\log \eta \sim \epsilon \log k$ , where  $\epsilon$  is the quantization exponent. In the table,  $\Delta$  is the aspect ratio of the dataset and  $\beta, \eta$  are the parameters from Theorem 5. Note that the relation  $\eta \leq \Delta$  always holds true. Moreover, under the manifold hypothesis the overall exponent  $\gamma$  is defined as  $\gamma = \epsilon + \rho$ .

| APPROACH                    | TIME COMPLEXITY                                                                                           | APPROXIMATION GUARANTEE                                                      |
|-----------------------------|-----------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|
| <i>Prior work</i>           |                                                                                                           |                                                                              |
| (Bachem et al., 2016a)      | $O(nD) + O(\epsilon^{-1} k^2 D \log k \epsilon^{-1})$                                                     | $O(\log k) \text{opt}_k + \epsilon \text{opt}_1$                             |
| (Bachem et al., 2018)       | $O(nD) + O(\epsilon^{-2} k^2 D^2 \log k)$                                                                 | $O(\log k) \text{opt}_k + \epsilon \text{opt}_1$                             |
| (Cohen-Addad et al., 2020a) | $O(n(D + \log n) \log(\Delta D))$<br>$+ O(\epsilon^{-1} k D^3 \log \Delta (n \log \Delta)^{O(\epsilon)})$ | $O(\epsilon^{-3} \log k) \text{opt}_k$                                       |
| (Charikar et al., 2023)     | $O(nD) + O(n \log n)$                                                                                     | $O(k^4 \log k) \text{opt}_k$                                                 |
| (Charikar et al., 2023)     | $O(nD) + O(n \log n)$<br>$+ O(\epsilon^{-2} k^5 D \log^2 k)$                                              | $O(\log k) \text{opt}_k$                                                     |
| (Draganov et al., 2024)     | $O(nD \log \log \eta) + O(\epsilon^{-4} k^2 D)$                                                           | $O(\log k) \text{opt}_k$                                                     |
| (Shah & Jaiswal, 2025)      | $O(nD) + O(\Delta^2 k^2 D)$                                                                               | $O(\log k) \text{opt}_k$                                                     |
| <i>Our results</i>          |                                                                                                           |                                                                              |
| Theorem 6                   | $O(nD) + O(mD \log k (k \log \eta)^{1+\epsilon})$                                                         | $O(\epsilon^{-2}) (\log k \text{opt}_k + k^{-\Omega(m/\beta)} \text{opt}_1)$ |
| Corollary 7                 | $O(nD) + O(\epsilon^{-1} \beta D (k \log \eta)^{1+\epsilon})$                                             | $O(\epsilon^{-2} \log k) \text{opt}_k$                                       |
| Corollary 8                 | $O(nD) + O(\epsilon^{-1} (k \log n)^{1+\gamma} D)$                                                        | $O(\epsilon^{-2} \log k) \text{opt}_k$                                       |

## Get In Touch!

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